

Scalability & Future Plan

Date	3 July 2026
Team ID	
Project Name	A Comprehensive Measure Of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently runs on a local Flask server and is not deployed to a cloud platform	Limits accessibility for multiple users.	High
2	The Linear Regression model may not accurately capture complex nonlinear relationships among development indicators.	May slightly affect prediction accuracy for some datasets.	Medium
3	The system relies on a static dataset and does not automatically incorporate newly published HDI data.	Predictions may not reflect the latest global development statistics.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local server.	Deploy the application on cloud platforms such as Render, Railway, or AWS to support concurrent users.
2	Data Storage	Uses a static CSV dataset.	Integrate a relational database such as MySQL or PostgreSQL to store and manage updated HDI records.
3	Performance	Uses a Linear Regression model for prediction.	Optimize preprocessing, implement caching, and evaluate advanced regression models for improved prediction accuracy and speed.
4	Security	Basic Flask application without user authentication.	Implement secure authentication, HTTPS, input validation, and role-based access control for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform (Render/AWS/Azure).	1-2 MonthS	Improve accessibility and availability for users.
Phase 2	Integrate a database to store prediction history and user information.	2-3 Months	Better data management and user experience.
Phase 3	Implement advanced Machine Learning models and Explainable AI techniques.	3-5 Months	Increase prediction accuracy and model transparency.
Phase 4	Develop an interactive dashboard with real-time analytics and visualizations.	5–6 Months	Enhance decision-making through richer insights and reporting.